iGEM 2026: Modelling Directions Report

ML/DL Strategy: ESM-2 Fine-tuning + Latent Bayesian Optimisation

Eton / Dry Lab / Substrate Predictor

Generative substrate design · Protein language models · Multi-objective optimisation

Executive Summary

This report proposes a two-stage machine learning pipeline for the iGEM 2026 Best Model award. **Stage 1** fine-tunes the ESM-2 protein language model on MEROPS cleavage data as a supervised multi-task classifier, producing a selectivity-aware embedding space for MMP-9, MMP-8, and HNE. **Stage 2** uses that fine-tuned embedding space as the search space for multi-objective Bayesian optimisation, generating *novel* 8-mer substrate candidates conditioned on a target selectivity profile. The top candidates are synthesised and validated experimentally by FRET cleavage assay, providing the model-driven experimental confirmation that iGEM judges reward above all other modelling criteria.

A secondary application (a Bayesian wound state classifier mapping biosensor output to clinical wound state probability distributions) is described as an auxiliary contribution that completes the clinical narrative of the project.

Core pipeline

Stage	Component	Method	Output
1a	ESM-2 encoder	Transfer learning from 650M-param PLM	Selectivity-tuned embeddings
1b	Attention analysis	Layer-wise relevance propagation	Position importance per protease
2a	GP surrogate	Gaussian process in tuned latent space	Selectivity landscape
2b	BO loop	EHVI multi-objective acquisition	Pareto-optimal novel sequences
2c	PSSM filter	Existing biochemical scorer	Defensibility gate
3	Wet lab	FRET cleavage assay	Experimental confirmation

Motivation and iGEM Judging Alignment

The 2026 judge handbook identifies four criteria for the Model special award:

1. How impressive is the modelling?
2. Did the model help the team understand a part, device, or system?
3. Were real measurements used to build or develop the model?
4. Does the approach provide a good example for others?

The Heidelberg 2025 exemplar (newly added to the 2026 handbook) establishes the current gold standard: a multi-layer computational framework where each layer is presented with full mathematical derivations, explicit parameter tables, and stated assumptions; and crucially, *the model predicted the optimal experimental condition, which wet lab then confirmed.*

The pipeline proposed here satisfies all four criteria:

- **Impressive**: protein language model fine-tuning + Bayesian optimisation in learned latent space is active academic research (BoTorch, ESM-2, EvoDiff lineage).
- **Informs design**: the BO output is a concrete, testable sequence, not a post-hoc rationalisation. The model decides what to synthesise.
- **Uses real measurements**: trained on MEROPS experimental cleavage data; validated against FRET-measured k_{cat}/K_m.
- **Reproducible**: full pipeline (fine-tuned weights, BO configuration, PSSM filter) can be re-run on any protease panel.

Stage 1: ESM-2 Fine-tuning as Selectivity-Aware Encoder

Motivation

The current substrate predictor uses VHSE8 encoding: eight hand-crafted physicochemical descriptors per amino acid position, producing a 64-dimensional feature vector. This is biologically motivated but limited; it encodes individual residue chemistry, not sequence context or evolutionary co-variation.

ESM-2 (Lin et al. 2022; Meta AI) is a transformer protein language model pre-trained on ∼250 million UniRef sequences. Its internal representations capture long-range residue interactions, conservation patterns, and functional site signatures learned from evolutionary data. For 8-mer cleavage windows, ESM-2 provides a far richer feature space than hand-crafted descriptors.

Architecture

A frozen or partially unfrozen ESM-2 trunk (we use the 650M parameter variant, `esm2_t33_650M_UR50D`) serves as a shared encoder. Three parallel linear classification heads sit on top, one per protease:

$$h_k = \sigma(W_k \cdot \mathrm{ESM2}(x) + b_k)\,, \quad k \in \{\text{MMP-9},\ \text{MMP-8},\ \text{HNE}\} \tag{1}$$

where x is the 8-mer sequence, $\mathrm{ESM2}(x) \in \mathbb{R}^{1280}$ is the mean-pooled per-residue representation from the final transformer layer, W_k and b_k are task-specific weights, and σ is the sigmoid function.

Training objective: multi-task binary cross-entropy with class-balanced sampling, identical to the existing LR pipeline. Fine-tuning proceeds in two phases: (i) head-only training with the ESM-2 trunk frozen (10 epochs); followed by (ii) full fine-tuning with a low learning rate applied to the trunk (10^{-5}, discriminative learning rates per layer). This prevents catastrophic forgetting while allowing the trunk to specialise.

Interpretability: Attention Analysis

Transformer attention maps from the fine-tuned model reveal which P-positions (P4–P1, P1'–P4') contribute most to each protease's classification. Layer-wise relevance propagation (LRP) or integrated gradients are applied to produce a per-position importance score $\alpha_{k,p}$ for protease k and position p.

This analysis produces a directly testable prediction: if the model assigns high $\alpha_{\text{MMP-9, P1}'}$ (position 5 in the 8-mer window), this independently confirms the known S1' pocket preference established by Kridel 2001. Positions where model attention and biochemical literature disagree are prioritised for point-mutant synthesis and experimental follow-up.

Outputs of Stage 1

- **Fine-tuned ESM-2 weights**: a selectivity-tuned encoder where the embedding space is explicitly shaped by protease cleavage data, not just evolutionary statistics.
- **Improved classification**: benchmarked against VHSE8 + LR baseline with the same train/test split and bootstrap CI protocol.
- **Attention importance maps**: per-position attribution per protease, with comparison to literature biochemistry.
- **Known-sequence validation**: `IPRTLTAA` and `AAPVSGIG` should score high on the correct head and near-zero on the others; serves as a sanity check on the fine-tuned model.

Stage 2: Latent Bayesian Optimisation for Generative Design

Why Bayesian Optimisation in the Fine-tuned Latent Space?

The fine-tuned ESM-2 embedding space is semantically structured: sequences with similar selectivity profiles cluster together, and the distance between embeddings correlates with differences in protease recognition. Running BO in this space is more efficient than searching raw sequence space ($20^8 \approx 25$ billion possible 8-mers) because the surrogate model can leverage the continuous, differentiable geometry of the embedding manifold.

Surrogate Model

A Gaussian process (GP) with Matérn-5/2 kernel is trained on the MEROPS training set to map ESM-2 embeddings $z \in \mathbb{R}^{1280}$ to the three-dimensional selectivity vector $[s_{\text{MMP-9}},\ s_{\text{MMP-8}},\ s_{\text{HNE}}]$. The GP provides not just a point estimate but a posterior predictive distribution $p(s \mid z)$, which is the uncertainty signal the acquisition function exploits.

For datasets of this size ($n \approx 1500$ total sequences), an exact GP is computationally tractable. For larger runs, a sparse variational GP (SVGP with $m = 200$ inducing points) scales linearly.

Acquisition Function and Multi-objective Search

The optimisation objective is Pareto-front discovery over three conflicting objectives: maximise $s_{\text{MMP-9}}$, minimise $s_{\text{MMP-8}}$, minimise s_{HNE}. The Expected Hypervolume Improvement (EHVI) acquisition function is used (as implemented in BoTorch):

$$\alpha_{\text{EHVI}}(z) = \mathbb{E}[\text{HVI}(\{s_{\text{MMP-9}}(z),\, -s_{\text{MMP-8}}(z),\, -s_{\text{HNE}}(z)\} \cup \mathcal{P})] \tag{2}$$

where $\mathcal{P}$ is the current Pareto front. The negative signs on MMP-8 and HNE convert the minimisation objectives to maximisation. The BO loop runs for 50 iterations; at each iteration the top 5 candidates by EHVI are decoded and scored with the PSSM filter.

Decoding Latent Candidates to Sequences

Candidates are proposed in continuous latent space and decoded to discrete sequences via nearest-neighbour lookup in the embedding space of a large (2M) background pool of random 8-mers scored by ESM-2. Alternatively, a lightweight sequence decoder (3-layer MLP trained to invert the encoder) can be used for finer-grained sequence recovery.

PSSM Filter

Before any candidate proceeds to wet lab consideration, it must satisfy:

- $\text{PSSM}_{\text{target}} > 0$ (positive log-odds for the intended protease)
- $\text{PSSM}_{\text{off-target}} < 2$ for both non-target proteases

This gate applies the existing mechanistic scorer as a biochemical prior, making the combined pipeline more defensible than the DL model alone. It is also the bridge between the existing substrate predictor work and the new generative pipeline.

Positive Control and Validation Strategy

The positive control is built into the design: if BO rediscovers `IPRTLTAA` or `AAPVSGIG` (sequences already in the training set), the pipeline's end-to-end consistency is confirmed. If it finds something better on the Pareto front, that is a genuine finding.

Wet lab validation: synthesise the top 2–3 novel candidates as FRET-labelled peptides (Dabcyl/EDANS or TAMRA/QSY-7) and measure k_{cat}/K_m for recombinant MMP-9 and HNE. A generated sequence with higher k_{cat}/K_m than the literature-derived `IPRTLTAA` would constitute strong validation of the generative model.

Auxiliary: Bayesian Wound State Classifier

Concept

The biosensor outputs a three-channel activity vector [MMP-9, MMP-8, HNE]. A downstream Bayesian classifier maps this to a probability distribution over wound states:

$$p(\text{state} \mid [\text{MMP-9}, \text{MMP-8}, \text{HNE}]) \quad \in \quad \{\text{healing},\ \text{chronic},\ \text{infected},\ \text{acute}\} \tag{3}$$

Data

Published wound fluid protease measurements (Trengove 1999, Ladwig 2002, Muller 2008, Watt 2009) are curated and harmonised via z-score normalisation within study. Total $n \approx 200$ patient samples across wound states. A MC-Dropout ensemble trained on this dataset provides both the point prediction and an epistemic uncertainty estimate; for example: "74 % probability chronic non-healing, high confidence" vs "53 % probability, low confidence: seek clinical review."

Role in the Project

This classifier is framed as the clinical application layer rather than the centrepiece model: *given* the biosensor works, here is how the output maps to actionable wound management decisions. ROC curves and optimal decision thresholds per wound state constitute a clear deliverable that judges can evaluate.

Integration with Existing Substrate Predictor

The VHSE8 + logistic regression + PSSM pipeline selects from *known* sequences in MEROPS. The new pipeline *designs* sequences outside the training distribution. They are complementary, not redundant:

Component	Role	Method
VHSE8 + LR + PSSM (existing)	Selects from known substrates	Feature engineering + LR
ESM-2 fine-tuned (Stage 1)	Richer selectivity encoder	Transfer learning
GP surrogate (Stage 2)	Selectivity landscape model	Gaussian process
EHVI BO (Stage 2)	Designs novel sequences	Multi-objective BO
PSSM filter (bridge)	Biochemical gate on BO output	Log-odds filter (existing)
Wound classifier (auxiliary)	Clinical decision support	MC-Dropout ensemble

The PSSM filter is the explicit bridge: the existing mechanistic model acts as a biochemical sanity check on DL outputs, making the combined pipeline more defensible than either component alone. The overall narrative is coherent: *we built a selectivity predictor, upgraded it to a language model, then used it to design sequences we couldn't have found by hand.*

Mentor Requirements

This project requires a mentor with experience in at least one of:

- **Protein sequence ML**: ESM-2 / protein language model fine-tuning, ProteinMPNN, EvoDiff, or equivalent. A lab working on de novo protein or peptide design would be ideal.
- **Bayesian optimisation for molecular design**: BoTorch, Dragonfly, or GP-based molecular property optimisation. Common in drug discovery computational groups.
- **Computational biology with ML focus**: any group applying deep learning to genomics, proteomics, or synthetic biology.

The software stack (PyTorch, HuggingFace `esm`, BoTorch, GPyTorch) is well-documented and widely used; a mentor does not need to be an ESM-2 specialist, but should be able to evaluate the modelling decisions critically.

Summary and Feasibility

Milestone	Complexity	Timeline	Deliverable
ESM-2 fine-tuning (Stage 1)	Medium	3–4 weeks	Fine-tuned weights; AUROC vs LR baseline
Attention analysis	Low	1 week	Position importance maps per protease
GP surrogate training	Medium	1–2 weeks	Surrogate model; predictive uncertainty
BO loop + PSSM filter	Medium	2–3 weeks	Pareto front; top 5 novel candidates
Wet lab FRET validation	High	4–6 weeks	k_{cat}/K_m for generated sequences
Wound classifier (aux.)	Low	1–2 weeks	ROC curves; decision thresholds
Total		**12–18 weeks**	Full Best Model submission

The single highest-risk component is wet lab FRET validation of the novel generated sequences. This risk is managed by (i) the PSSM gate ensuring biochemical plausibility before synthesis, and (ii) the positive control (rediscovery of `IPRTLTAA`) confirming end-to-end pipeline consistency before novel sequences are committed. The computational components (Stages 1 and 2) are independently valuable and reportable even if wet lab validation is limited.